Aryan Singh

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Education

Technische Universität Hamburg Oct 2024 – Mar 2027
ICS – Secure and Dependable IT Systems *Hamburg, Germany*
Master of Science

Indian Institute of Information Technology Aug 2019 – May 2023
Electronics and Communication Engineering *Bhopal, India*
Bachelor of Technology *GPA: 8.93/10*

Experience

Working Student – Enterprise Architecture Nov 2024 – Present
Eppendorf SE *Hamburg, Germany*

- Built an AI-powered internal knowledge assistant using Langdock, MCP, and enterprise documentation to help employees quickly find application information.
- Developed Python automation for SAP LeanIX data collection and maintenance using REST and GraphQL APIs, reducing manual data-management effort.
- Engineered a Python-based parser to automate SAP LeanIX permission management, reducing manual configuration effort.
- Automated the migration of GDPR processing activity records from SharePoint to LeanIX using a custom Python script, improving data consolidation and compliance tracking.

Full Stack Developer – Consultant Jul 2023 – Sep 2024
AST LLC *Remote*

- Maintained and developed the TaaS team's web portal using Spring Boot and React.
- Led the frontend upgrade for the web portal, updating legacy codebase to use modern React 18+ features such as Functional Components, Hooks etc.
- Automated complex project onboarding process by developing a web-based wizard, saving hundreds of manual hours.
- Introduced functionality to publish and lock a Test Run in real-time from the web portal, which is now utilized by over 29 team members, significantly reducing the time and efforts of the support team.

Full Stack Developer Jan 2023 – Jul 2023
D-WebBox *Mumbai, India*

- Led the frontend for the FOR ALL COMMUNITY application for Great Place To Work®, built entirely from scratch for Android and iOS platforms.
- Designed and developed entire backend system for Prelux Asian Paints from scratch, a feature complete dashboard system for management of contractor meets and events.
- Implemented complex peer scoring mechanism logic at scale for GenWE application.
- Performed code reviews and enhanced the performance of several React Native applications.

React Native Intern Jun 2022 – Jan 2023
D-WebBox *Remote*

- Enhanced iOS specific features and UX of FIKAA, an application built for investing in the stock market.
- Built the Amyraa cross-platform e-commerce application for Android and iOS using React Native.
- Improved iOS features and resolved launch-critical issues across production React Native applications.

Frontend Development Intern Oct 2021 – Jun 2022
TravOkarma *Remote*

- Improved the performance of the User and Creator dashboard section by optimising Network calls and refactoring UI code.
- Designed and developed several core features such as Profile and Blogs section on the app from scratch.

Software Development Intern Oct 2020 – Oct 2021
Remassis *Remote*

- Developed a cross-platform iOS/Android Online Consultation App in React Native
- Built backend microservices with Node.js, GraphQL and MySQL, including real-time video calls using WebRTC.

Open Source

Ancient Beast: Contributed to an open-source turn-based strategy game by implementing core gameplay features and UI enhancements using TypeScript and Less/CSS.

Twenty CRM: Contributed frontend enhancements to an open-source CRM built with NestJS, React, and TypeScript, including composite field support in the Excel Import Wizard.

Projects

Code Churn in Vulnerability and Bug Commits: An Empirical Analysis *(Projektarbeit, Institute of Software Security, TUHH)*: Built a custom code churn extraction pipeline to analyze 126,000+ security and bug-fix commits across Java and C/C++ projects using four curated datasets.

School ERP for JMRD, Datia: Designed and built a full-stack School ERP system with attendance, administration, notice board, and doubt management features using React Native, Node.js, and MongoDB.

MovieFinder: iOS app built in Swift and SwiftUI that retrieves movie information from the OMDb API and features an offline watchlist using CoreData.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, Swift, HTML/CSS

Frameworks: Spring Boot, React, React Native, Node.js, Express.js, Flask, SwiftUI

Databases & APIs: MySQL, MongoDB, REST APIs, GraphQL

Cloud & Enterprise: AWS, Docker, LeanIX, SAP

AI & Automation: MCP (Model Context Protocol), LLM Applications, AI Agents, RAG

Developer Tools: Git, GitHub Actions, VS Code, Eclipse, Xcode, Vim, LaTeX

Achievements/Awards

Rising Star Award at AST LLC: Awarded for valuable contributions to Automation Cloud Tester (ACT), Web Portal and Support Tools development

MMVY Scholarship: Received full tuition fee from state Government for the entire bachelors program

IEEE Hacksagon: Finalist at Hacksagon conducted by IEEE at IIIT Gwalior

Languages

German: B2

English: IELTS 8.5/9 (CEFR C2)

Hindi: Native